MAINTENANCE MANUAL FOR DAMAGED CONNECTOR ON QUADFOIL

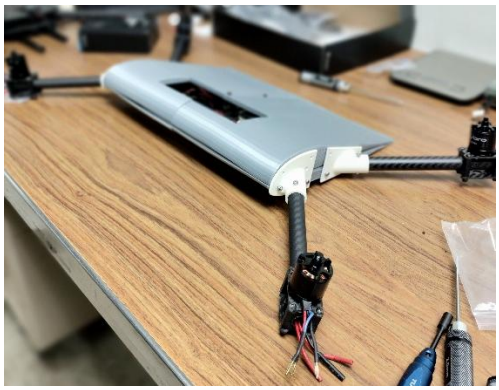
1. **Prepare the Aircraft for Maintenance**

   Stabilize the Quadfoil and remove all power sources before handling structural components.

i. Disconnect the battery and ensure propellers are removed.

ii. Place the Quadfoil on a padded surface to avoid additional stress on arms.

iii. Inspect the aircraft visually to confirm only the bracket is damaged and not the arm or center body.



1a. Battery disconnected

1b. Motor-ESC connector disconnected

2. **Access the Damaged Connector Bracket**

   Let's say that the rear right connector is broken as shown in image 2b.

i. Remove the allen fasteners indicated by red circle in the image 2a with 3mm allen driver and 6mm hex nut wrench.

ii. Gently remove the motor arm and connector assembly from the center body as shown in image 2c.

2a. Rear right connector attached to center body

2b. Broken connector/ bracket

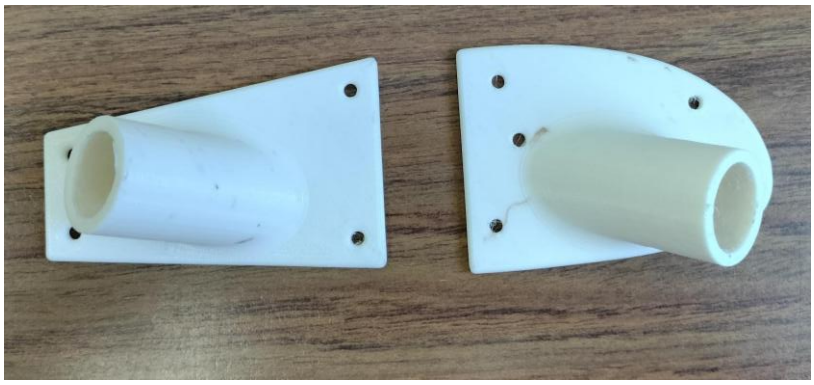
2c. Arm motor assembly removed from center body

3. Reprint the damaged part

i. 3D Print the connector which is broken by using the STL Files.

ii. All replacement parts are provided as STL files. Users must slice these files on their own 3D printer to generate machine-specific G-code.

iii. Sand the parts lightly using low-grit sandpaper (180 grit) to avoid removing excess material.



3a. Right front and back connector

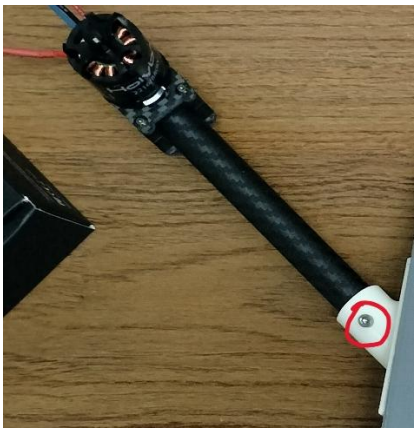
4. Install the new connector on the arm

i. Insert the arm into the connector so that the hole in the outer cylindrical section of the connector aligns with the hole in the carbon-fiber arm, as shown in figure 4a.

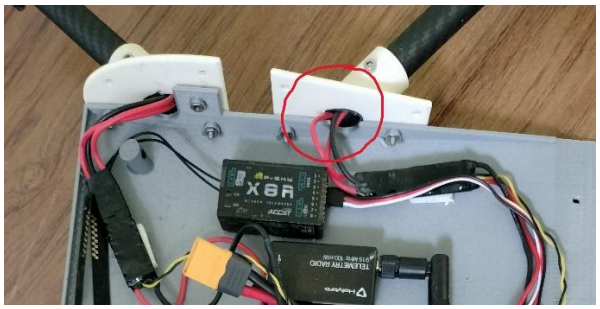
ii. Secure the arm to the connector using an M2.5 × 16 mm bolt and matching nut, tightening them until the arm is firmly clamped in place, as shown in figure 4b.

iii. Route the ESC wires through the arm before fastening, ensuring the wires exit near the motor end and are positioned close to the motor connectors, as shown in figure 4c and figure 4d.

4a. Match the holes



4b. Secure the arm on connector



4c. Wire routing from center body through connector 4d. Wire routed to the exit near motor

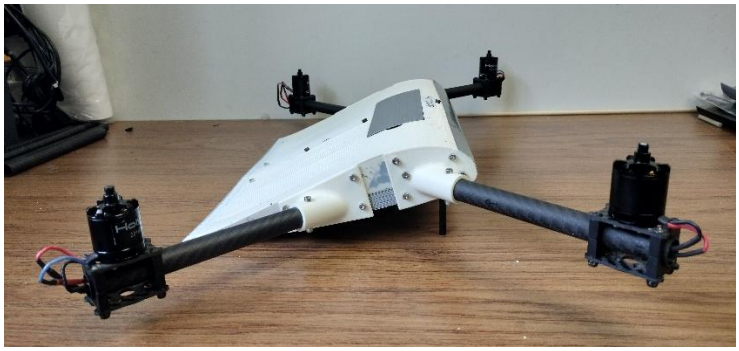
5. Attach the connector to the center body

i. Secure the bracket to the fuselage mounting points

ii. Align bolt holes precisely; do not force misaligned hardware.

iii. Insert M 2.5 x 12 bolts; tighten until snug as seen in image 5a.

iv. Verify no gaps exist between bracket and fuselage.



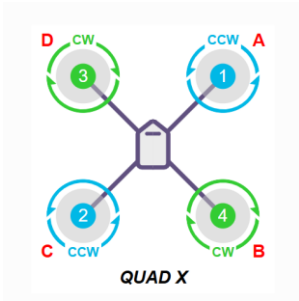
5a. Connector attached to center body

6. Perform structural checks and install propellers

   i. Apply light manual bending to the arm to confirm bracket rigidity.

   ii. Install battery, power on the aircraft and apply some throttle to check for any vibration in the arm due to motor rotation.

   iii. If everything looks good, disconnect the battery and install CW & CCW propellers to their respective motors as seen in image 6a. Happy flying!



6a. Motor direction for X frame type quadcopter 6b. Happy flying!